EE 413 · COMMUNICATION SYSTEMS II

# Digital Communications

*Lecture notes for the whole course: how a waveform becomes a finite alphabet of symbols, how those symbols are carried across a channel that adds noise to everything, how a receiver decides which one was sent, and how often it is wrong.*

| Covers | Level | Assumed background |
| --- | --- | --- |
| Chapters 1 and 2 | Undergraduate | Fourier analysis, probability, random processes |

### How to read these notes

Every chapter builds on the one before it. Within a chapter, each idea arrives in the same order: a picture, a definition, an equation, a short derivation, a worked example, and a warning about the mistake that is easiest to make. Worked examples use five headings — Given, Find, Method, Solution, Check. Do the Check step yourself before reading it.

Two conventions apply everywhere and both are worth fixing now, because getting either wrong costs a factor of two that nothing on the page reveals. Noise is white and Gaussian with **two-sided** power spectral density $N_0/2$ watts per hertz. The Gaussian tail is $Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty e^{-t^2/2}\, dt = \frac{1}{2}\,\mathrm{erfc}(x/\sqrt{2})$. Energy and power are normalised, with $R = 1\ \Omega$.

The contents below carries a third column. An entry such as **PS CH7.2.1** points into the course textbook, Proakis and Salehi, *Fundamentals of Communication Systems*, second edition, where the same material is developed at length. The **PS** marker is what tells the two apart, and it is not decorative: these notes reach information theory in chapter 6 and the textbook develops it in chapter 12, while the textbook's own chapter 12 heading is nowhere near what a bare number would suggest.

and the raised cosine.

CHAPTER 1

# The transition from analog to digital

*A continuous waveform becomes a bit stream in three steps: sampling makes it discrete in time, quantization makes it discrete in amplitude, and encoding replaces each amplitude by a word of bits. Only the first of the three can be undone, and this chapter is largely about what that means.*

## 1.1 Impulse-train sampling

Sampling every $T_s$ seconds is multiplication by a train of impulses. Write $f_s = 1/T_s$ for the sampling frequency.

THE IDEAL SAMPLED SIGNAL

$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$$

$$g_\delta(t) = g(t)\, p(t) = \sum_{n=-\infty}^{\infty} g(nT_s)\, \delta(t - nT_s)$$

The second line uses the sampling property $g(t)\delta(t-t_0) = g(t_0)\delta(t-t_0)$, which turns the product under the sum into a number.

What makes this worth writing is that $g_\delta$ is a continuous-time signal, not a sequence, so it has a Fourier transform. A product in time is a convolution in frequency, and the transform of an impulse train is another impulse train:

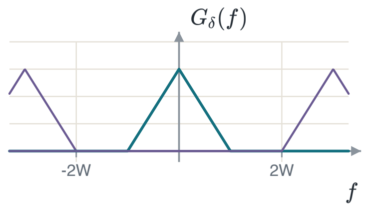
$$P(f) = \frac{1}{T_s} \sum_{n=-\infty}^{\infty} \delta(f - nf_s) = f_s \sum_{n=-\infty}^{\infty} \delta(f - nf_s)$$

The coefficient is the Fourier-series coefficient of $p(t)$, and it is the same for every harmonic: $a_k = \frac{1}{T_s}\int_{-T_s/2}^{T_s/2} \delta(t) e^{-j2\pi k f_0 t}\, dt = \frac{1}{T_s}$, by the sifting property. Convolving $G$ with that train gives the result the rest of the chapter rests on.

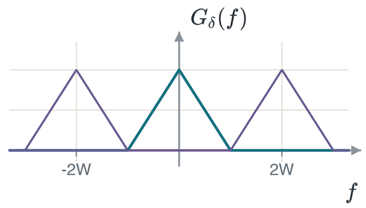
SAMPLING REPLICATES THE SPECTRUM

$$G_\delta(f) = f_s \sum_{n=-\infty}^{\infty} G(f - nf_s)$$

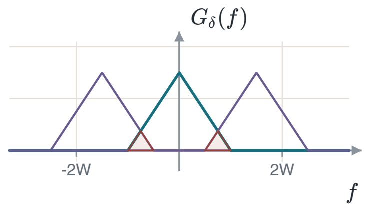
Sampling copies the spectrum to every multiple of $f_s$ and scales it by $f_s$. Nothing is lost provided the copies do not overlap.




$f_s > 2W$: a gap between the replicas.




$f_s = 2W$: they touch and do not overlap.




$f_s < 2W$: they overlap, and the overlap cannot be undone.

ALIASING

In the third case a high frequency of the message has been added to a low frequency of a replica. No filter can separate a sum into its parts, so the original signal cannot be recovered from its samples — not by a better filter and not by more computation. In practice a real signal is never strictly bandlimited, so a **guard band** $f_g$ is left between the edge of the message and the edge of the first replica: $f_s = 2W + f_g$, with an anti-aliasing filter removing whatever lies above $W$ before the sampler sees it.

## 1.2 The sampling theorem and reconstruction

SAMPLING THEOREM

If $G(f) = 0$ for $|f| \geq W$ and $f_s \geq 2W$, then $g(t)$ is determined completely by its samples $g(nT_s)$ and can be recovered from them exactly. If $f_s < 2W$, aliasing occurs and the recovery fails. The rate $2W$ is the **Nyquist rate** and $T_s = 1/(2W)$ the **Nyquist interval**.

Recovery means keeping the copy at the origin and rejecting every other one, which is a lowpass filter. At $f_s = 2W$:

THE RECONSTRUCTION FILTER AND ITS IMPULSE RESPONSE

$$H_{\text{LPF}}(f) = \begin{cases} \dfrac{1}{2W}, & |f| \leq W \\ 0, & \text{otherwise} \end{cases}$$

$$h_{\text{LPF}}(t) = \int_{-W}^{W} \frac{1}{2W} e^{j2\pi f t}\, df = \frac{\sin(2\pi W t)}{2\pi W t} = \operatorname{sinc}(2Wt)$$

The convention used throughout is $\operatorname{sinc}(x) = \sin(\pi x)/(\pi x)$, so that sinc is one at zero and zero at every other integer.

THE GAIN IS NOT ONE

Sampling multiplied the spectrum by $f_s = 2W$, so the filter has to divide it back. A reconstruction filter of unit gain returns a signal $2W$ times too large, and nothing in a plot of the spectrum *shape* reveals it.

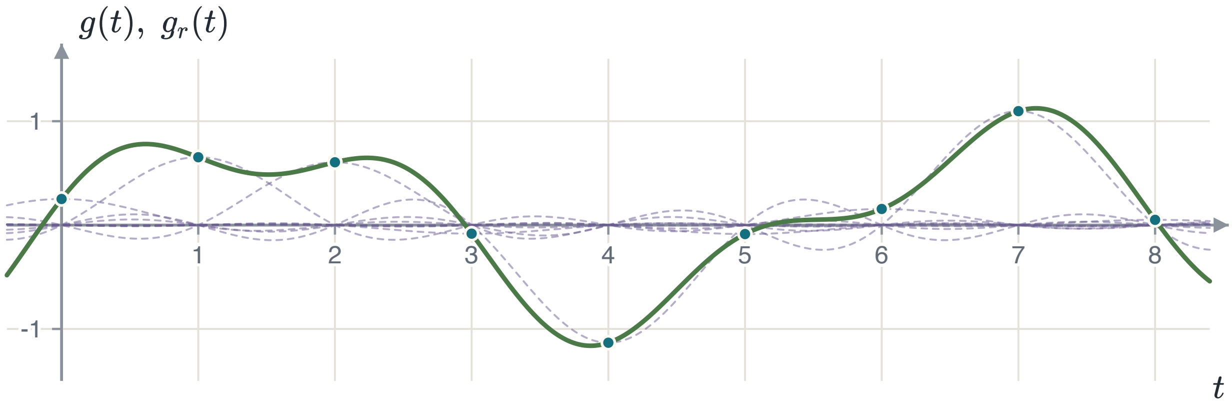
Filtering in frequency is convolution in time. Convolving the impulse train with $h_{\mathrm{LPF}}$ and using the sifting property once more gives the interpolation formula.

INTERPOLATION

$$g_r(t) = \sum_{n=-\infty}^{\infty} g(nT_s)\,\mathrm{sinc}\big(2W(t - nT_s)\big)$$

$$g_r(t) = \sum_{n=-\infty}^{\infty} g\left(\frac{n}{2W}\right)\mathrm{sinc}(2Wt - n) \quad \text{at } f_s = 2W$$

At $t = kT_s$ every term vanishes except the one with $n = k$, because sinc is zero at every non-zero integer. So $g_r(kT_s) = g(kT_s)$ exactly — and, since the filter passes the whole message and nothing else, the two agree everywhere and not only at the instants.




Each sample contributes one shifted sinc scaled by its own value. Their sum, drawn heavy, passes through every sample because each sinc is zero at all the other sampling instants.

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EXAMPLE 1.1 — THREE SAMPLING RATES

GIVEN A signal $x(t)$ bandlimited to $W = 40$ kHz.

FIND (a) the Nyquist rate; (b) the rate with a $10$ kHz guard band; (c) the Nyquist rate of $y(t) = x(t)\cos(80000\pi t)$.

METHOD Read (a) and (b) off the geometry of the replicas. For (c) find the bandwidth of $y$ first: the rate follows from that, not from the bandwidth of $x$.

SOLUTION (a) $f_s = 2W = 80$ kHz. (b) $f_s = 2W + f_g = 90$ kHz. (c) The carrier is at $f_c = 40$ kHz, and multiplication shifts the spectrum both ways and halves it: $Y(f) = \frac{1}{2}X(f - 40\mathrm{k}) + \frac{1}{2}X(f + 40\mathrm{k})$. The result occupies $|f| < 80$ kHz, so $f_s = 160$ kHz.

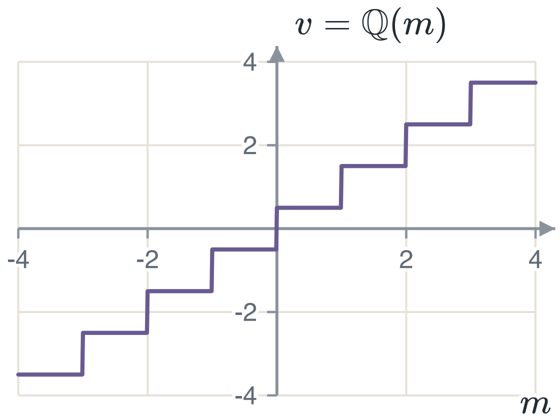
CHECK The highest frequency of $y$ is $f_c + W = 80$ kHz and $2 \times 80 = 160$ kHz. Doubling the carrier would double the answer while leaving the width of $Y$ unchanged: the rate follows the highest frequency present, not the width of the message.

THE TRAP IN PART (C)

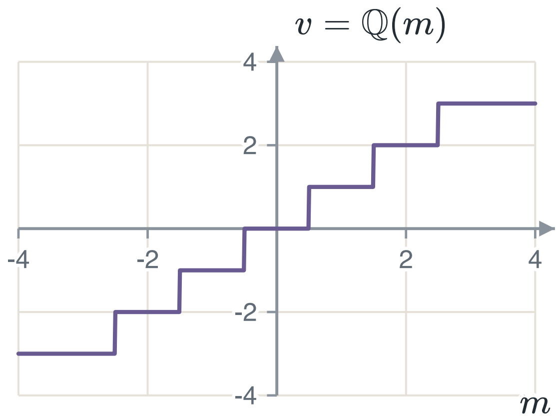
Answering $80$ kHz is answering for $x$, not for $y$. Modulation moved the message up the frequency axis and the sampler has to keep up with where it went.

## 1.3 Quantization

Quantization replaces a sample amplitude by the nearest member of a finite set of $L$ **representation levels**. In a **uniform** quantizer the spacing $\Delta$ between consecutive levels is the same everywhere. The two families differ in what happens at zero: a **mid-rise** quantizer puts a decision boundary there, so no output level is zero, while a **mid-tread** quantizer puts a level there, so a small input is quantized to exactly zero. Speech coders use mid-tread, because a silent input should be reproduced as silence rather than as a $\pm\Delta/2$ chatter.




Mid-rise, $L = 8$, $\Delta = 1$. A boundary at the origin.




Mid-tread, $L = 8$, $\Delta = 1$. A level at the origin.

A quantizer is a partition of the input range into $L$ regions $\mathcal{J}_k$ with one output value $v_k$ per region. Two conditions make it optimal, and they refer to each other, which is why they are applied by turns rather than solved at once.

1. Each boundary is the **midpoint** of the two levels it separates, $m_k = \frac{1}{2}(v_{k-1} + v_k)$. Given the levels, this minimises the error, because it sends every input to the nearer level.
2. Each level is the **centroid** of its own region, $v_k = E[M \mid M \in \mathcal{J}_k]$. Given the boundaries, this minimises the mean square error inside the region.

A uniform quantizer satisfies the first by construction. It satisfies the second only when the input is uniformly distributed, which is the deeper reason the uniform quantizer is optimal for a uniform source and for no other.

## 1.4 Quantization noise and the signal-to-noise ratio

The quantization error is $Q = M - \mathbb{Q}(M)$. Because every input is sent to the nearer level, it can never exceed half a step, so $-\Delta/2 \le q \le \Delta/2$. If $\Delta$ is small enough that the density of $M$ is nearly flat across one region, the error is as likely to fall anywhere in that interval as anywhere else.

THE UNIFORM ERROR MODEL

$$f_Q(q) = \frac{1}{\Delta}, \qquad -\frac{\Delta}{2} \le q \le \frac{\Delta}{2}$$

$$E[Q^2] = \int_{-\Delta/2}^{\Delta/2} q^2 \frac{1}{\Delta}\, dq = \frac{\Delta^2}{12}$$

With a zero-mean input the error has zero mean too, so this mean square is also the variance: the power of the quantization noise is its variance.

Substituting $\Delta = 2m_{\max}/L$ and $L = 2^R$ puts the result in the form the rest of the chapter uses, and dividing the signal power by it gives the figure of merit.

SIGNAL-TO-QUANTIZATION-NOISE RATIO

$$E[Q^2] = \frac{1}{12}\left(\frac{2m_{\max}}{L}\right)^2 = \frac{m_{\max}^2}{3 \cdot 2^{2R}}$$

$$\mathrm{SQNR} = \frac{P_M}{E[Q^2]} = \frac{3P_M}{m_{\max}^2} 2^{2R}$$

$$\mathrm{SQNR}\ [\mathrm{dB}] = \underbrace{10 \log_{10} \frac{3P_M}{m_{\max}^2}}_{\alpha} +\ 6.02R$$

Every extra bit per sample adds $20 \log_{10} 2 = 6.02$ dB. Doubling the level count halves the step, which quarters the error power, and a factor of four is $6.02$ dB.

WHAT $\alpha$ COSTS

The intercept is negative for every signal that does not fill the quantizer range. A full-scale sinusoid has $P_M = m_{\max}^2/2$ and $\alpha = 10 \log_{10} 1.5 \approx 1.76$ dB; a sinusoid reaching a quarter of the range has $\alpha = -10.28$ dB, a loss of $12.04$ dB — exactly two bits. That calculation is the argument for companding, and its size is the size of the argument.

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EXAMPLE 1.2 — A FULL-SCALE SINUSOID

GIVEN $m(t) = 5 \cos t$ through a uniform quantizer spanning its full range.

FIND The step size and the SQNR at $R = 3$ and $R = 4$ bits per sample.

METHOD Average power from Parseval, step size from $\Delta = 2m_{\max}/L$, then $\alpha + 6.02R$.

SOLUTION The coefficients are $a_{\pm 1} = 5/2$, so $P_M = 2(5/2)^2 = 12.5$ and $m_{\max} = 5$. At $R = 3$: $\Delta = 2(5)/8 = 1.25$ V and $\mathrm{SQNR} = 1.76 + 18.06 = 19.82$ dB. At $R = 4$: $\Delta = 0.625$ V and $\mathrm{SQNR} = 1.76 + 24.08 = 25.84$ dB.

CHECK The two differ by $6.02$ dB, one bit. Independently, $E[Q^2] = \Delta^2/12 = 0.1302$ and $10 \log_{10}(12.5/0.1302) = 19.82$ dB.

WHAT THE FORMULA DOES NOT KNOW

Quantize this waveform and measure the error it actually makes, and the answers are $19.09$ dB and $25.31$ dB — about $0.7$ dB and $0.5$ dB below the formula. The uniform model assumes the error is equally likely anywhere in half a step, and a sinusoid spends most of its time near its peaks, where the error is largest. The gap halves with every extra bit: $0.27$ dB at $R = 6$ and $0.14$ dB at $R = 8$. That is what "$\Delta$ small enough" means, and knowing the size of the gap is the difference between using a model and believing it.

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EXAMPLE 1.3 — WHEN THE MODEL BREAKS

GIVEN A zero-mean stationary Gaussian source with $S_X(f) = 2$ for $|f| < 100$ Hz, sampled at the Nyquist rate, each sample through a five-level quantizer with outputs $-30, -10, 0, 10, 30$ and boundaries at $-40, -20, 20, 40$.

FIND The SQNR of the scheme.

METHOD The signal power is the area under the spectral density. The noise power must be integrated region by region, because this quantizer is neither uniform nor fine.

SOLUTION $P_X = \int_{-100}^{100} 2\, df = 400$, and since the mean is zero this is also $\sigma_X^2$. Splitting $P_Q = \int (x - \mathbb{Q}(x))^2 f_X(x)\, dx$ at the four boundaries gives $7.98 + 46.36 + 79.50 + 46.36 + 7.98 = 188.18$, so $\mathrm{SQNR} = 10 \log_{10}(400/188.18) = 3.27$ dB.

CHECK The central region alone contributes $79.50$, and it holds the $68\%$ of the mass with $|X| < 20$ and an error of up to $20$. That is where a five-level quantizer spends its error, and it is why the answer is a few decibels rather than a few tens.

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WHY $\Delta^2/12$ WOULD HAVE LIED

The quantizer is coarse and its outer regions are unbounded, so the error there is not bounded by half a step at all: a sample at $x = 120$ is quantized to $30$ and the error is $90$. Applying the uniform model with $\Delta = 20$ predicts $E[Q^2] = 33.3$ and an SQNR of $10.8$ dB, three times too optimistic. The model is a small-step model, and this is what its failure looks like.

## 1.5 Non-uniform quantization

Speech spends most of its time at small amplitudes and only occasionally reaches its peak. A uniform quantizer gives the same absolute step to a whisper and to a shout, so the whisper is quantized far more coarsely in proportion to itself. Relaxing the requirement that the regions be of equal length allows the distortion to be reduced at the same level count — narrow regions where the density is high and wide ones where it is low, which is what the centroid condition asks for and what a uniform quantizer cannot give.

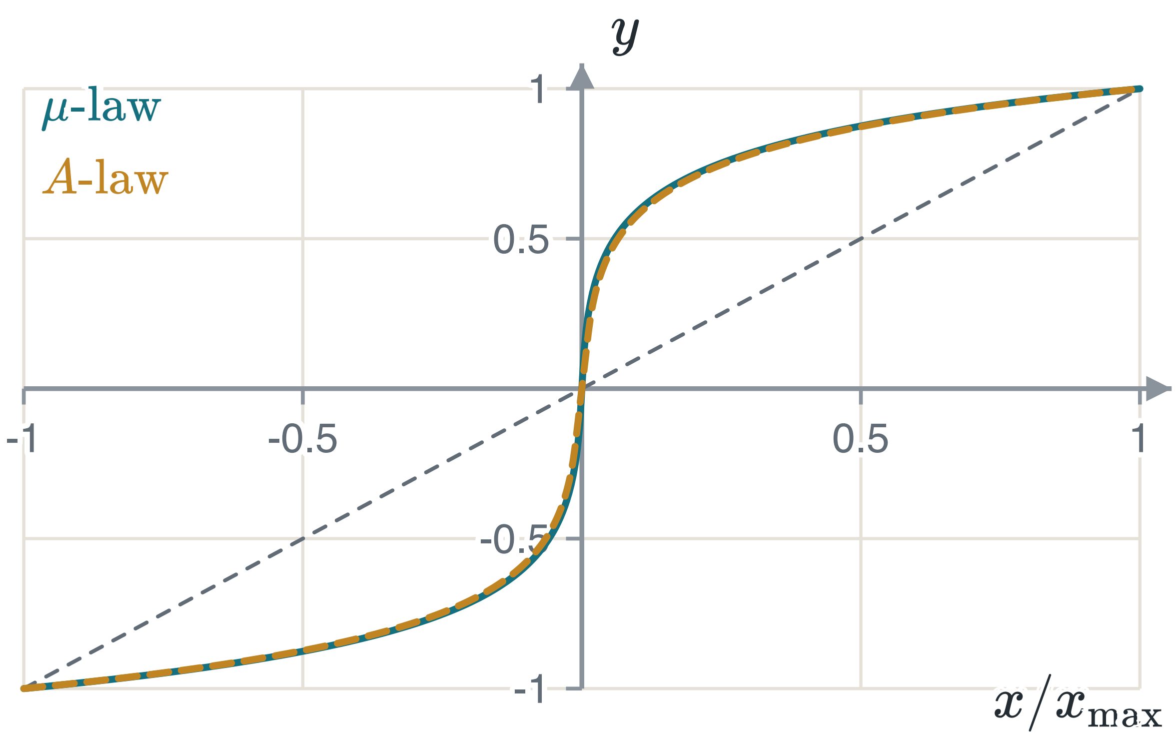
Rather than build a non-uniform quantizer, the signal is compressed by a memoryless non-linearity, quantized **uniformly**, and expanded at the receiver by the inverse. Compressing plus expanding is **companding**, and it is how the non-uniform quantizer is built out of a uniform one.

THE TWO STANDARD COMPRESSORS

$$y = \frac{\ln(1+\mu|x|)}{\ln(1+\mu)}\,\mathrm{sgn}(x), \qquad |x| \le 1$$

$$y = \begin{cases} \dfrac{A|x|}{1+\ln A}\,\mathrm{sgn}(x), & 0 \le |x| \le 1/A \\ \dfrac{1+\ln(A|x|)}{1+\ln A}\,\mathrm{sgn}(x), & 1/A < |x| \le 1 \end{cases}$$

Both are normalised so that $x = \pm 1$ maps to $y = \pm 1$. The standard parameters are $\mu = 255$ and $A = 87.6$ ; on the same speech through the same eight-bit uniform quantizer the two give signal-to-noise ratios within a hundredth of a decibel of one another. The difference between them is regional rather than technical.

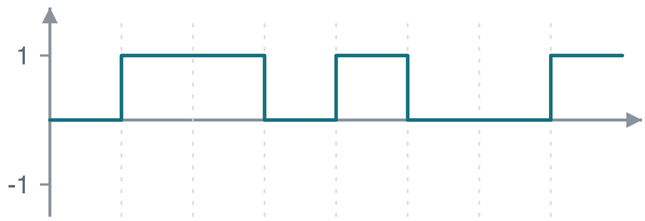



The two compressor characteristics against the identity, which is what no companding would give. Both spend most of their output range on the smallest tenth of the input.

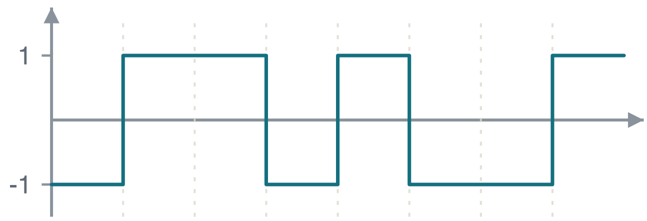
## 1.6 Encoding, line codes and pulse code modulation

With $L = 2^R$ levels each sample needs $R$ bits, and at $f_s$ samples per second the stream leaves the encoder at $R_b = Rf_s$ bits per second. **Natural binary coding** assigns $0$ to $L-1$ to the levels in increasing order; **Gray coding** assigns the words so that adjacent levels differ in exactly one bit. The second is worth the trouble because a channel error almost always moves the decision to a neighbouring level, and under natural binary that can flip several bits at once — level $7$ is $0111$ and level $8$ is $1000$, four bits apart, where the Gray words $0100$ and $1100$ are one.

A **line code** is the rule that turns those bits into a waveform. Unipolar NRZ is on-off signalling: simplest, and it carries a DC level that makes the waveform droop through any AC-coupled stage. Polar NRZ is antipodal signalling and has no DC component for balanced data, which is why it is the shape Chapter 2 analyses. Both go quiet during a long run of identical bits, and a receiver recovering its clock from the waveform then has nothing to lock to; Manchester forces a transition in the middle of every bit, so the clock is always recoverable and there is never a DC component, paid for with twice the bandwidth.




Unipolar NRZ.




Polar NRZ, carrying the same eight bits $01101001$.

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EXAMPLE 1.4 — A COMPLETE PCM STREAM

GIVEN $m(t) = 8|\operatorname{sinc}(t-2)|$, so $0 \le m(t) \le 8$, sampled every $T_s = 0.6$ s and quantized by an eight-level uniform quantizer covering $[0, 8]$.

FIND The step size, the levels, the code words for $t = 0, 0.6, \ldots, 3.6$, and the bit rate.

METHOD $\Delta$ from the range and the level count; each sample by direct evaluation; each word from the tread the sample falls in.

SOLUTION $\Delta = 8/8 = 1$ V and the levels sit at $0.5, 1.5, \ldots, 7.5$. The samples are $0,\ 1.73,\ 1.87,\ 7.48,\ 6.05,\ 0,\ 1.51$, giving the words $000\ 001\ 001\ 111\ 110\ 000\ 001$. With $R = 3$ and $f_s = 1/0.6 = 1.667$, $R_b = 5$ bit/s and $T_b = T_s/3 = 0.2$ s.

CHECK The sample at $t = 3$ is $8|\operatorname{sinc}(1)| = 0$ exactly, because $\operatorname{sinc}$ vanishes at every non-zero integer — the same fact the interpolation formula rests on, met again at the other end of the chapter.

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## 1.7 Summary

| RESULT | STATEMENT | ANCHOR |
|---|---|---|
| Replication | $G_\delta(f) = f_s \sum_n G(f - nf_s)$ | PS CH7.1.1 |
| Sampling theorem | $f_s \geq 2W$ for a message bandlimited to $W$ | PS CH7.1.1 |
| Reconstruction | $g_r(t) = \sum_n g(nT_s)\,\mathrm{sinc}(2Wt - n)$, filter gain $1/(2W)$ | PS CH7.1.1 |
| Step size | $\Delta = 2m_{\max}/L,\ \ L = 2^R$ | PS CH7.2.1 |
| Error power | $E[Q^2] = \Delta^2/12$, when the step is small | PS CH7.2.1 |
| Signal-to-noise | $\mathrm{SQNR}\ [\mathrm{dB}] = \alpha + 6.02R$ | PS CH7.2.1 |
| Bit rate | $R_b = Rf_s$ | PS CH7.3, 7.4.1 |

Sampling is reversible and quantization is not. Everything after this chapter takes the bit stream as given and asks what the channel does to it.

CHAPTER 2

# Baseband transmission of digital signals

*Chapter 1 produced a stream of bits. Now each bit is sent as a waveform, a channel adds noise to it, and a receiver has to say which waveform was sent. This chapter answers three questions in order: what filter the receiver should use, where the decision threshold should go, and what happens when the channel is too narrow and the pulses start to overlap.*

## 2.1 The matched filter

Over one bit interval the receiver sees $x(t) = g(t) + w(t)$, the waveform plus white Gaussian noise of two-sided density $N_0/2$. It passes this through a filter and takes one sample at the end of the interval. Because the filter is linear, its output splits the same way, into a signal part $g_0(t) = g * h$ and a noise part $n(t) = w * h$.

The quantity worth maximising is the signal at the sampling instant against the noise power that accompanies it.

THE PEAK PULSE SIGNAL-TO-NOISE RATIO

$$(\mathrm{SNR})_o = \frac{|g_0(T)|^2}{E[n^2(t)]} = \frac{\left|\int_{-\infty}^{\infty} G(f)H(f)e^{j2\pi fT}df\right|^2}{\frac{N_0}{2}\int_{-\infty}^{\infty}|H(f)|^2 df}$$

Scaling $H$ up multiplies the top and the bottom by the same factor, so the answer cannot be "make $H$ large". What is being chosen is the shape of $H$.

Schwarz's inequality says that $\left|\int \phi_1\phi_2\right|^2 \leq \int |\phi_1|^2 \int |\phi_2|^2$, with equality only when $\phi_1 = k\phi_2^*$. Taking $\phi_1 = H(f)$ and $\phi_2 = G(f)e^{j2\pi fT}$, the factor $\int |H|^2$ cancels between the top and the bottom and a bound appears that contains no $H$ at all.

THE BOUND, AND THE FILTER THAT REACHES IT

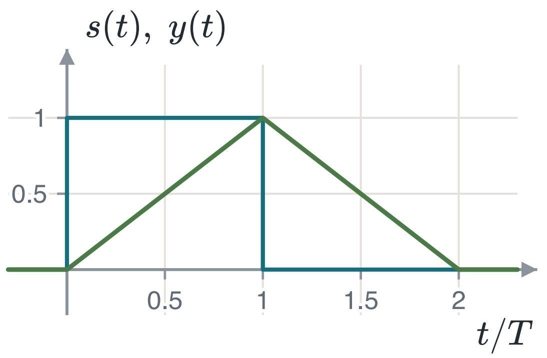
$$(\mathrm{SNR})_o \le \frac{2}{N_0}\int |G(f)|^2 df = \frac{2E}{N_0}$$

$$H(f) = k\,G^*(f)e^{-j2\pi fT} \quad \Longleftrightarrow \quad h_{\mathrm{opt}}(t) = k\,g(T-t)$$

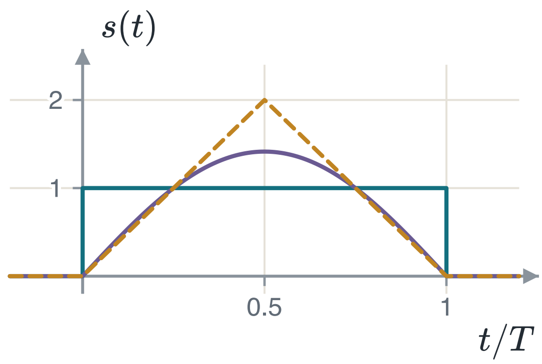
The optimum filter is the transmitted waveform reversed in time and shifted into the interval. It is called the **matched filter**.

TWO THINGS THIS SAYS

The best achievable ratio is $2E/N_0$: it depends on the **energy** of the pulse and on the noise density, and on nothing else about the pulse. Two completely different waveforms of the same energy perform identically. And the constant $k$ cancels, so the filter is fixed only up to a gain.




A rectangular pulse and the output of the filter matched to it. The output peaks at exactly $t = T$, and its peak is the energy of the pulse.




Three pulses of the same energy. Against white Gaussian noise a matched-filter receiver performs identically on all three.

## 2.2 From waveform to number

Both waveforms of a binary baseband system are multiples of one shape, so a single unit-energy function carries both. With $\psi(t) = 1/\sqrt{T_b}$ on the bit interval, polar NRZ is $s_m(t) = s_m\psi(t)$ with $s_0 = -A\sqrt{T_b}$ and $s_1 = +A\sqrt{T_b}$. Both carry energy $E_b = A^2T_b$, so $A\sqrt{T_b} = \sqrt{E_b}$ and the two waveforms have become two *numbers*, $\pm\sqrt{E_b}$, on one axis.

The demodulator can be built two ways. The **matched filter** convolves with $\psi(T_b - t)$ and samples at $T_b$. The **correlator** multiplies by $\psi(t)$ and integrates over the interval. At the sampling instant the two produce the same number, because convolving with a reversed function and evaluating at the end of the interval *is* correlating over it.

THE DECISION STATISTIC

$$y = \int_0^{T_b} x(\tau)\psi(\tau)\,d\tau = s_m\underbrace{\int_0^{T_b}\psi^2}_{=1} + \underbrace{\int_0^{T_b} w\psi}_{n} = s_m + n$$

They agree at $t = T_b$ and nowhere else, which is one more reason the sampling instant matters.

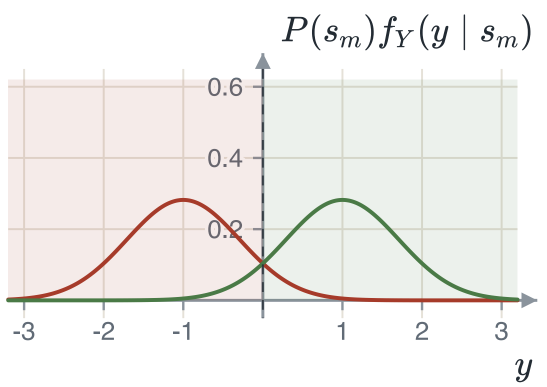
## 2.3 The decision and its error probability

The noise term is a projection of a Gaussian process onto a fixed function, so it is a Gaussian random variable. Its variance follows from $E[w(\tau)w(u)] = \frac{N_0}{2}\delta(\tau - u)$ and the sifting property, and the unit energy of $\psi$ makes the answer independent of the interval length.

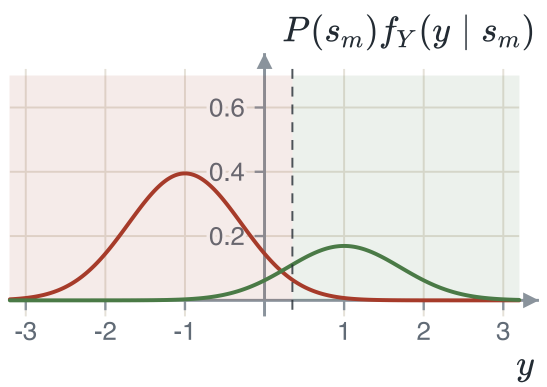
> WHAT THE DETECTOR IS GIVEN
>
> $$\sigma_n^2 = \frac{N_0}{2}\int_0^{T_b}\psi^2(\tau)d\tau = \frac{N_0}{2}$$
>
> $$y = s_m + n \sim \mathcal{N}\left(s_m, \frac{N_0}{2}\right)$$
>
> The two-sided convention arrives here unchanged. Reading the density as one-sided makes every error probability in the course $3$ dB too optimistic, and nothing in the algebra shows it.

The detector decides "1" when $y > \lambda$. Writing the average error as a sum of two integrals, differentiating with respect to $\lambda$ by the Leibniz rule and setting the derivative to zero gives a condition that is easy to read: the threshold sits where the two densities, each weighted by its prior, cross.

> THE OPTIMAL THRESHOLD
>
> $$P(s_1)f_Y(\lambda \mid s_1) = P(s_0)f_Y(\lambda \mid s_0)$$
>
> $$\lambda_{\text{opt}} = \frac{N_0}{4\sqrt{E_b}}\ln\frac{P(s_0)}{P(s_1)}$$
>
> Equal priors put it midway. If $s_0$ is the more likely symbol the threshold moves in the positive direction, enlarging the region that decides $s_0$ — which is the right way round.




Equal priors: the threshold sits midway and the shaded decision regions are symmetric.




$P(s_0) = 0.7$: the more likely symbol's curve is taller and the crossing has moved towards the less likely one.

With equal priors the threshold is at the origin and both conditional errors are the same Gaussian tail, using $P(Y > y) = Q\left(\frac{y-\mu}{\sigma}\right)$ and $Q(-x) = 1 - Q(x)$.

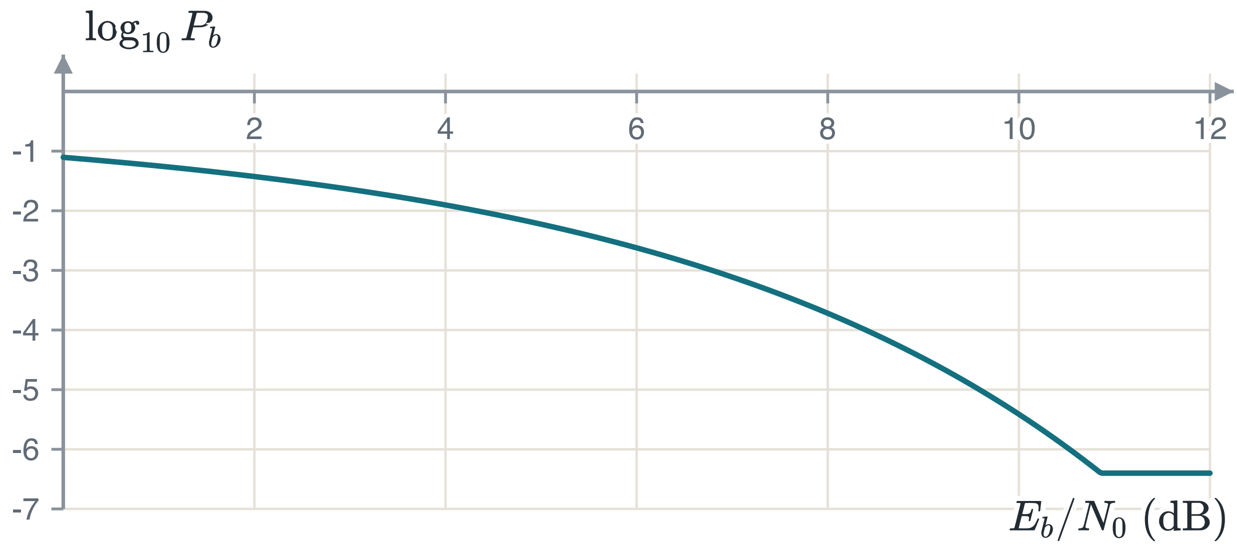
THE RESULT OF THE CHAPTER

$$P_b = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$$

It depends on $E_b/N_0$ and on nothing else — not on the amplitude, not on the bit duration, not on the pulse shape. Doubling the amplitude and quartering the duration change neither the energy per bit nor the answer.

THE FACTOR OF TWO THAT LIVES HERE

$Q\left(\sqrt{E_b/N_0}\right)$, the same expression without the two, is the error probability of on-off or orthogonal signalling, and it is $3$ dB worse. Which one applies depends on whether the two waveforms are opposites or merely different. For on-off signalling half the bits carry no energy at all, so the average energy per bit is half the peak — and forgetting that is how on-off comes out looking as good as antipodal.




Bit error probability against $E_b/N_0$. Past about $8$ dB every extra decibel is worth roughly an order of magnitude in error rate.

EXAMPLE 2.1 — UNEQUAL PRIORS

GIVEN A binary PAM system with correlator output $y = \pm\sqrt{E_b} + n$, where $P(s_1) = 0.3$, $E_b = 1$ and $N_0 = 0.1$.

FIND The optimal threshold and the average error probability there.

METHOD Threshold from the log-ratio of the priors; each conditional error from a Gaussian tail; average by weighting.

SOLUTION $\lambda = \frac{0.1}{4} \ln \frac{0.7}{0.3} = 0.0212$ and $\sigma = \sqrt{0.05} = 0.2236$. Then $P(\text{err} \mid s_0) = Q(4.567) = 2.475 \times 10^{-6}$ and $P(\text{err} \mid s_1) = Q(4.377) = 6.005 \times 10^{-6}$, so $P_e = 0.7(2.475) + 0.3(6.005)$ in units of $10^{-6}$, giving $3.534 \times 10^{-6}$.

CHECK Leaving the threshold at zero would give $Q(\sqrt{20}) = 3.872 \times 10^{-6}$. Moving it improves the answer by about nine per cent — a real gain and a small one, which is what a shift of a tenth of a standard deviation buys.

## 2.4 Intersymbol interference

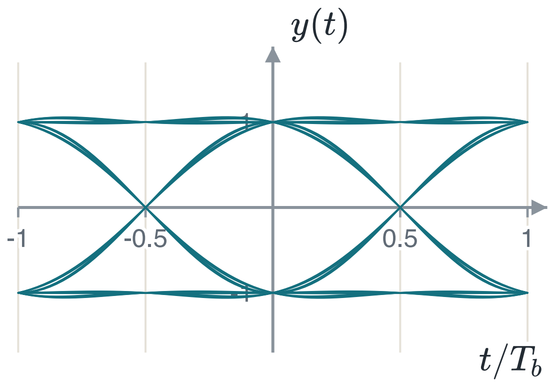
Everything above assumed an ideal channel. A real channel is bandlimited, and a bandlimited channel spreads each pulse in time, so a pulse meant for one interval leaks into the next. Writing the received signal as a train of the overall pulse $p$ and sampling at $t_i = iT_b$ separates the term that was meant to be there from the ones that were not.

WHERE THE INTERFERENCE COMES FROM

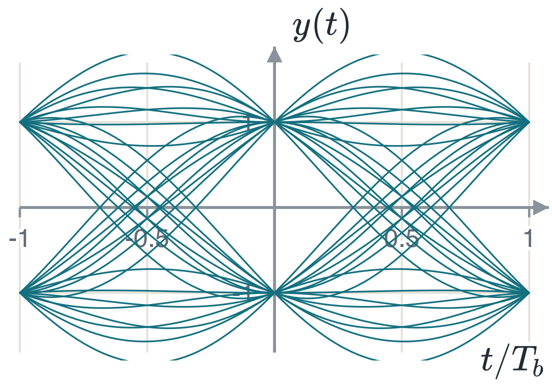
$$y(t_i) = \underbrace{\mu a_i}_{\text{wanted}} + \underbrace{\mu \sum_{k \neq i} a_k \, p\big((i-k)T_b\big)}_{\text{intersymbol interference}} + n(t_i)$$

The middle term is not noise. It is caused by the data itself, and raising the transmit power raises it by the same factor, so more power does not help. At high signal-to-noise ratio it is the only thing limiting the system.

The **eye pattern** is how this is seen rather than calculated. Cut the received waveform into bit-length pieces, draw them all on top of each other synchronised to the clock, and four measurements are readable at once: the height of the opening is the margin over noise, its width is how far the sampling instant may drift, the slope at the crossings is the sensitivity to a timing error, and the spread of the crossings is the timing jitter. When the opening closes there is no sampling instant at which every bit pattern decides correctly, and no threshold placement helps.




A wide-open eye: a pulse that decays quickly. Every trace passes close to $\pm 1$ at the centre.




A slowly decaying pulse. The opening has narrowed and the crossings have scattered, with the noise unchanged.

## 2.5 Nyquist's criterion and the raised cosine

The interference vanishes exactly when the overall pulse is zero at every sampling instant but its own. Sampling $p$ at those instants and transforming turns that requirement into a statement about the spectrum.

NYQUIST'S CRITERION FOR DISTORTIONLESS TRANSMISSION

$$p\big((i-k)T_b\big) = \begin{cases} 1, & i = k \\ 0, & i \neq k \end{cases}$$

$$R_b \sum_n P(f - nR_b) = 1, \qquad R_b = \frac{1}{T_b}$$

In words: the replicas of the pulse spectrum, spaced by the symbol rate, must add to a constant. The simplest spectrum that does it is a rectangle of width $2W$ with $W = R_b/2$, the **Nyquist bandwidth**, whose pulse is $\mathrm{sinc}(2Wt)$.

THE RATE A BANDWIDTH SUPPORTS

A channel of bandwidth $W$ carries at most $2W$ symbols per second with no interference. This is the counterpart of the sampling theorem, reached from the other end of the same argument.

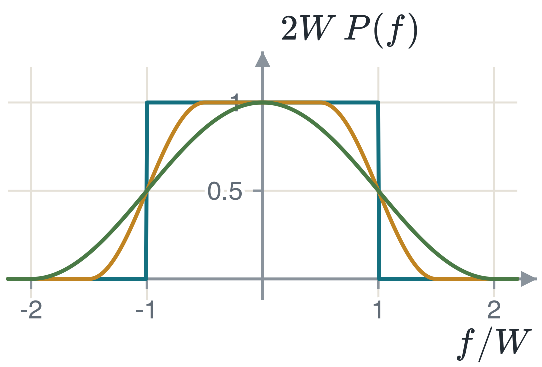
The ideal Nyquist channel cannot be built: its spectrum has vertical edges, and its pulse decays only as $1/t$, so a small timing error lets a long tail of neighbours contribute at once. The fix is to widen the spectrum and taper its edges while keeping the tiling property.

THE RAISED COSINE

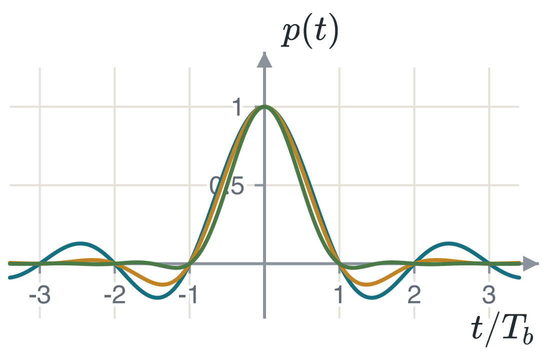
$$P(f) = \begin{cases} \dfrac{1}{2W}, & 0 \le |f| \le f_1 \\ \dfrac{1}{4W}\left[1 - \sin\dfrac{\pi(|f| - W)}{2W - 2f_1}\right], & f_1 \le |f| < 2W - f_1 \\ 0, & \text{otherwise} \end{cases}$$

$$p(t) = \mathrm{sinc}(2Wt)\,\frac{\cos(2\pi\alpha W t)}{1 - 16\alpha^2 W^2 t^2}, \qquad \alpha = 1 - \frac{f_1}{W}$$

The first factor keeps the zero crossings at $t = iT_b$, so the criterion still holds; the second decays as $1/t^2$. The price is bandwidth: $B_T = (1+\alpha)W$, an excess of $\alpha W$ over Nyquist.




The spectrum at $\alpha = 0, 0.5$ and $1$. All three tile the axis at spacing $2W$, so all three give zero interference.




Their pulses. All three vanish at every non-zero multiple of $T_b$; the tails differ by orders of magnitude.

EXAMPLE 2.2 — FITTING A RATE INTO A CHANNEL

| | |
|---|---|
| GIVEN | A channel of bandwidth $48$ kHz is to carry $64$ kbit/s with raised-cosine shaping. |
| FIND | The Nyquist bandwidth and the largest roll-off that fits. |
| METHOD | $W = R_b/2$, then $(1+\alpha)W \le B$. |
| SOLUTION | $W = 32$ kHz, and $(1+\alpha)(32) \le 48$ gives $\alpha \le 0.5$. At $\alpha = 0.5$ the signal uses the whole $48$ kHz, with an excess of $16$ kHz over the Nyquist bandwidth. |
| CHECK | At $\alpha = 0$ the signal would need only $32$ kHz and would also fit — and would be a bad choice, because its pulse decays as $1/t$ and its filter cannot be built. Using the whole channel buys a pulse decaying as $1/t^3$ and a filter that exists. The spectral efficiency is $64/48 = 1.33$ bit/s/Hz against the theoretical $2$. |

## 2.6 Summary

| RESULT | STATEMENT | ANCHOR |
|---|---|---|
| Matched filter | $h_{\text{opt}}(t) = g(T-t)$ | PS CH8.3.2 |
| What it achieves | $(\text{SNR})_o = 2E/N_0$, independent of the pulse shape | PS CH8.3.2 |
| Decision statistic | $y = s_m + n,\ n \sim \mathcal{N}(0, N_0/2)$ | PS CH8.3.1 |
| Optimal threshold | $\lambda = \frac{N_0}{4\sqrt{E_b}} \ln \frac{P(s_0)}{P(s_1)}$ | PS CH8.3.3 |
| Antipodal error | $P_b = Q\left(\sqrt{2E_b/N_0}\right)$ | PS CH8.3.3 |
| On-off error | $P_b = Q\left(\sqrt{E_b/N_0}\right)$, three decibels worse | PS CH8.3.3 |
| Nyquist criterion | $R_b \sum_n P(f - nR_b) = 1$ | PS CH10.3.1 |
| Raised cosine | $B_T = (1+\alpha)W$ | PS CH10.3.1 |

Two waveforms have become two points on a line, and the error probability has turned out to depend on the distance between them. Chapter 3 asks what happens when there are more than two waveforms, and finds that the same picture works with more axes.